

DDA5002 Optimization

Lecture 8 IP Modeling

Totally Unimodular

Midterm Review

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Outline

- 1 IP Modeling Review
- 2 Selection
- 3 Facility Location Problem
- 4 LP Relaxation of IP
- 5 Min Cost Network Flow and Totally Unimodular
- 6 Midterm Review

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General "If ... then ..." Conditions

If $f(x_1, x_2, \dots, x_n) \geq a$, then $g(x_1, x_2, \dots, x_n) \geq b$.

$$\leq 0 + M(\epsilon)$$

$$\geq 0 - M(\epsilon)$$

$$\begin{cases} f(x_1, x_2, \dots, x_n) \leq a + My \\ g(x_1, x_2, \dots, x_n) \geq b - M(1 - y) \\ y \in \{0, 1\} \end{cases}$$

General Disjunctive Constraints

Either $f(x_1, x_2, \dots, x_n) \leq a$ or $g(x_1, x_2, \dots, x_n) \leq b$



If $f(x_1, x_2, \dots, x_n) > a$, then $g(x_1, x_2, \dots, x_n) \leq b$



$$\begin{cases} f(x_1, x_2, \dots, x_n) \leq a + My \\ g(x_1, x_2, \dots, x_n) \leq b + M(1 - y) \\ y \in \{0, 1\} \end{cases}$$

K-out-of-N Must Hold

Out of the N following constraints, K must be satisfied:

$$f_1(x_1, \dots, x_n) \leq b_1$$

$$f_2(x_1, \dots, x_n) \leq b_2$$

$$\vdots$$

$$f_N(x_1, \dots, x_n) \leq b_N$$

K-out-of-N Must Hold

$$f_1(x_1, \dots, x_n) \leq b_1 + M(1 - y_1)$$

$$f_2(x_1, \dots, x_n) \leq b_2 + M(1 - y_2)$$

$$\vdots$$

$$f_N(x_1, \dots, x_n) \leq b_N + M(1 - y_N)$$

$$\sum_{j=1}^N y_j = K$$

$$y_j \in \{0, 1\}, \quad j = 1, 2, \dots, N.$$

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Discrete variables

$$x \in \{2, 3, 5, 7, 10, 12\}$$

$$x \in \{2.5, 6.7, 8.3, 11.75\}$$

- A discrete variable has a domain of the form

$$x \in \{a_1, a_2, \dots, a_K\}$$

- If $a_1, a_2, \dots, a_K$ are contiguous integers, we can model as

$$x \in \{1, 2, 3, 4, \dots, K\} \quad \hookrightarrow \quad a_1 \leq x \leq a_K, x \in \mathbb{Z}$$

- Otherwise we can model as

$$x = \sum_{k=1}^K a_k y_k, \quad \sum_{k=1}^K y_k = 1, y_k \in \{0, 1\}, k = 1, \dots, K$$

$$x \in \{a_1, a_2, \dots, a_k\}$$

$\Downarrow$
 y_1

$\Downarrow$
 y_2

$\Downarrow$
 y_k

$$y_i = \begin{cases} 1 & \text{if select } a_i \\ 0 & \text{o.w.} \end{cases}$$

$$\left\{ \begin{array}{l} x = \sum_{i=1}^k a_i y_i = a_1 y_1 + a_2 y_2 + \dots + a_k y_k \\ \sum_{i=1}^k y_i = 1 \\ y_i \in \{0, 1\}, \forall i=1, \dots, k \end{array} \right.$$

Semicontinuous variables

- A semicontinuous variable has a domain of the form

$$x \in [l_1, u_1] \cup [l_2, u_2] \cup \cdots \cup [l_K, u_K]$$

where $l_1 \leq u_1 \leq \cdots \leq l_K \leq u_K$

- One modeling approach for this system is

$$x = \sum_{k=1}^K z_k$$

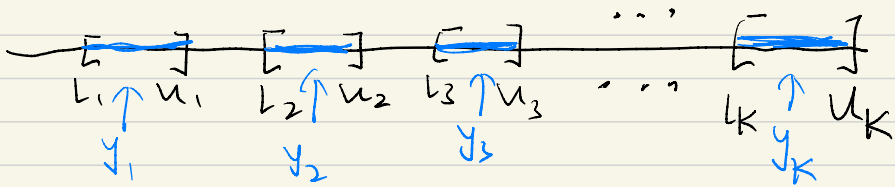
$$l_k y_k \leq z_k \leq u_k y_k \quad k = 1, \dots, K$$

$$\sum_{k=1}^K y_k = 1$$

$$y_k \in \{0, 1\} \quad k = 1, \dots, K$$

eg. $x \in [1, 2] \vee [3, 4] \vee [5, 6] \vee [7, 8]$

$$\left\{ \begin{array}{l} 1 \leq x \leq 2 \\ 3 \leq x \leq 4 \\ 5 \leq x \leq 6 \\ 7 \leq x \leq 8 \end{array} \right. \rightarrow \text{WRONG} \\ \text{Infeasible}$$



$$y_i = \begin{cases} 1 & \text{if } l_i \leq x \leq u_i \\ 0 & \text{o.w.} \end{cases}$$

$$\left\{ \begin{array}{l} l_i y_i \leq x \leq u_i y_i, \quad \forall i=1, \dots, K \\ \sum_{i=1}^K y_i = 1 \\ y_i \in \{0, 1\}, \quad \forall i=1, \dots, K \end{array} \right. \rightarrow \text{WRONG}$$

$\rightarrow$ If $y_i = 0$, $0 \leq x \leq 0 \Rightarrow x = 0$

$$l_i y_i \leq z_i \leq u_i y_i, \quad \forall i=1, \dots, K$$

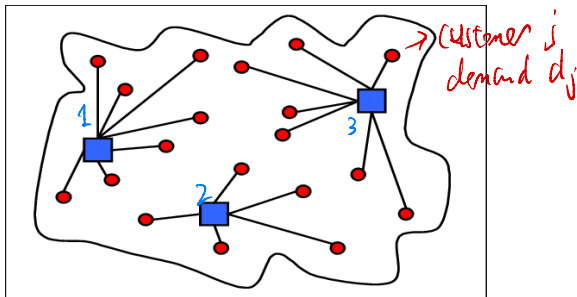
$$x = \sum_{i=1}^K z_i = z_1 + z_2 + \dots + z_K$$

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Example: Warehouse/Facility Location Problem

- There are n warehouses available for use.
- Opening warehouse i has a fixed operating cost f_i .
- There are m customers.
- Customer j has demand d_j . It costs c_{ij} to ship one unit of product from warehouse i to customer j .
- The objective is to satisfy all customers' demands, while minimizing the total costs (operating + shipment).



Formulation

Decision variables:

$$y_i = \begin{cases} 1 & \text{if open warehouse } i \\ 0 & \text{o.w.} \end{cases}$$

- y_i : whether to open warehouse i or not ($y_i \in \{0, 1\}$)
- x_{ij} : how many units to ship from warehouse i to customer j

Objective function:

$$\min \underbrace{\sum_{i=1}^n f_i y_i}_{\text{opening cost}} + \underbrace{\sum_{i,j} c_{ij} x_{ij}}_{\text{shipping cost}}$$

Constraints:

- Need to satisfy every customer's demand: $\underline{\sum_{i=1}^n x_{ij} \geq d_j}$
- A location can provide supply only if it is opened :

$$0 \leq x_{ij} \leq d_j y_i$$

A warehouse can deliver products to customers only if the warehouse is open.

$$\begin{cases} \text{If } y_i = 1, & \text{then } x_{ij} \geq 0 \\ \text{If } y_i = 0, & \text{then } x_{ij} = 0 \end{cases}$$

$$0 \leq x_{ij} \leq d_j y_i, \quad \forall i, \forall j$$

$$y_i = 1 \Rightarrow 0 \leq x_{ij} \leq d_j$$

$$y_i = 0 \Rightarrow 0 \leq x_{ij} \leq 0 \Rightarrow x_{ij} = 0$$

Take-home exercise,

A warehouse need to supply at least 2 products if it is open.

Formulation

Therefore we can formulate the warehouse/facility location problem as follows:

$$\begin{array}{ll}\min & \sum_{i=1}^n f_i y_i + \sum_{i,j} c_{ij} x_{ij} \\ \text{s.t.} & \sum_{i=1}^n x_{ij} \geq d_j, \quad \forall j \\ & x_{ij} \leq d_j y_i, \quad \forall i, j \\ & x_{ij} \geq 0, \quad \forall i, j \\ & y_i \in \{0, 1\}, \quad \forall i\end{array}$$

It is a mixed integer linear program.

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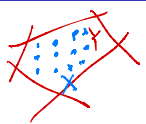
LP Relaxation of an IP

Integer Program (IP):

$$\begin{aligned} V^{\text{IP}} = \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \\ & x \in \mathbb{Z}^n \end{aligned}$$

(Note: A blue 'X' is drawn next to the constraints, and a blue circle highlights the integer requirement $x \in \mathbb{Z}^n$.)

LP Relaxation of IP:


$$\begin{aligned} V^{\text{LP}} = \min \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

(Note: A red circle highlights the constraints $Ax = b$ and $x \geq 0$, and a red 'Y' is drawn to the right.)

- LP relaxation removes the integer requirement.
- The feasible region of the LP relaxation is a superset of the feasible region of the IP.

$$\text{lower bound} \rightarrow V^{\text{LP}} \leq V^{\text{IP}}$$

(Note: The inequality $X \subseteq Y$ is written above the main inequality.)

LP Relaxation as a Bound for IP

Theorem (LP Relaxation as a Bound for IP)

- ① *For a maximization problem, the optimal value of the LP relaxation provides an upper bound for the optimal value of the IP.*
- ② *For a minimization problem, the optimal value of the LP relaxation provides a lower bound for the optimal value of the IP.*

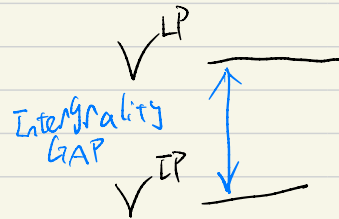
The difference between the optimal value of the LP and the IP is called the **integrality gap**

- For maximization problems, the integrality gap is $v^{LP} - v^{IP}$
- For minimization problems, the integrality gap is $v^{IP} - v^{LP}$

$$\begin{aligned}
 & \text{(IP)} \\
 V^{\text{IP}} = \max. & \quad C^T x \\
 \text{s.t.} & \quad Ax = b \\
 & \quad x \geq 0 \\
 & \quad x \in \mathbb{Z}
 \end{aligned}$$

$$V^{\text{LP}} \geq V^{\text{IP}}$$

$$\begin{aligned}
 & \text{LP Relaxation} \\
 V^{\text{LP}} = \max. & \quad C^T x \\
 \text{s.t.} & \quad Ax = b \\
 & \quad x \geq 0
 \end{aligned}$$



$$x^0 \in X, x^0 \in \mathbb{Z} \quad V^0 = C^T x^0$$

$$\text{Compare } V^{\text{IP}}, V^{\text{LP}}, V^0$$

$$V^0 \leq V^{\text{IP}} \leq V^{\text{LP}}$$

$$\Rightarrow 0 \leq \underbrace{V^{\text{IP}} - V^0}_{\text{Small}} \leq \underbrace{V^{\text{LP}} - V^0}_{\text{Known Small}}$$

Use LP Relaxation as a Bound

Consider a maximization problem, suppose we solved the LP relaxation and get the optimal value is v^{LP} (with some fractional solution).

Then we use some method to obtain a feasible integral solution (e.g., by rounding the solution to the LP relaxation) and find the objective value is v^0 .

Then the difference between the obtained solution and the optimal integral solution (whose objective value is v^{IP}) satisfies:

$$0 \leq v^{IP} - v^0 \leq v^{LP} - v^0$$

That is, one can use the LP relaxation solution to construct a bound on how good a certain integral solution is.

IP $\rightarrow x^*$ optimal

$$\max. C^T x$$

$$\text{s.t. } Ax = b$$

$$x \geq 0$$

$$x \in Z$$

LP relaxation $\rightarrow x^{**}$ optimal

$$\max. C^T x$$

$$\text{s.t. } Ax = b$$

$$x \geq 0$$

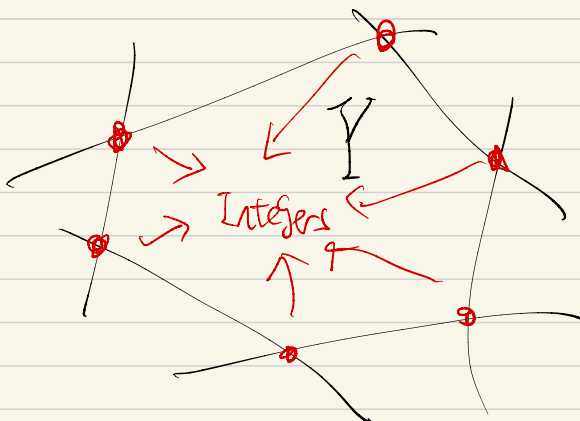
$$x \in Y$$

$$X \subseteq Y$$

Thm: If $x^{**} \in Z$, then $x^* = x^{**}$.

x^{**} is also optimal for IP.

$$\text{Proof: } \left. \begin{array}{l} x^{**} \in Z \\ x^{**} \in Y \end{array} \right\} \Rightarrow x^{**} \in X$$



Some Good Cases

There are some cases where solving the LP relaxation can give one very good solution

Theorem

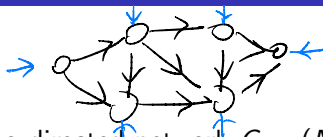
If the optimal solution to the LP relaxation is integral, then the solution must be optimal to the IP problem.

- Question: when?
- Answer: Remember for LP, there must exist an optimal solution that is a basic feasible solution. If we can guarantee that every BFS is integral, then the LP must have an integer optimal solution.
- Example: Min-cost Network Flow.

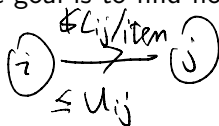
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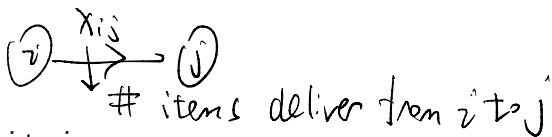
Minimum Cost Network Flow Model



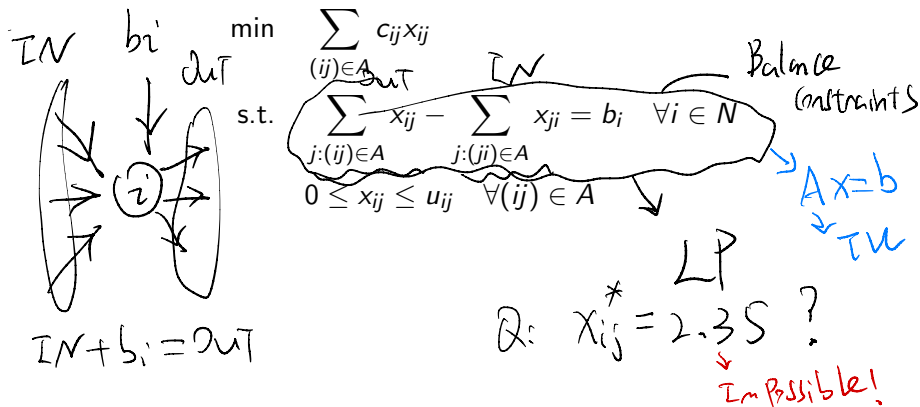
- We are given a directed network $G = (N, A)$ with a set of nodes N and a set of arcs A .
- Each node $i \in N$ has an associated “supply” b_i . If $b_i > 0$ the node is a supply node, if $b_i < 0$ it is a demand node. We will assume that the network is balanced, i.e. $\sum_{i \in N} b_i = 0$.
- Each arc $(i, j) \in A$ has an associated cost c_{ij} and capacity u_{ij} .
- A flow on this network is a set of values on the arcs that obey capacities and satisfy flow conservation at the nodes.
- The goal is to find flows on each arc with minimum cost.



Minimum Cost Network Flow Problem

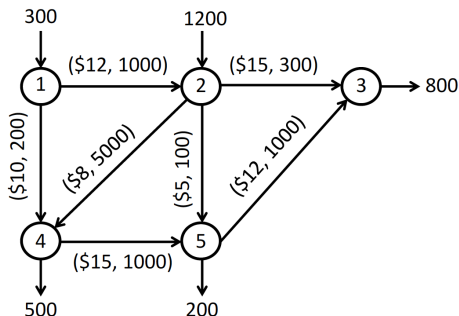


Let x_{ij} denote the flow from i to j



Example

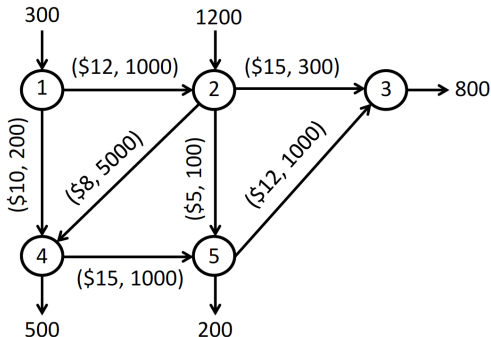
- A company has two manufacturing plants and 3 distribution centers (nodes)
- The plants and DCs are connected by transportation channels (arcs)
- The production amount and demand at each node is shown
- The capacity and cost per unit for each channel (arc) is shown
- Find the cheapest way to move units from plants to DCs



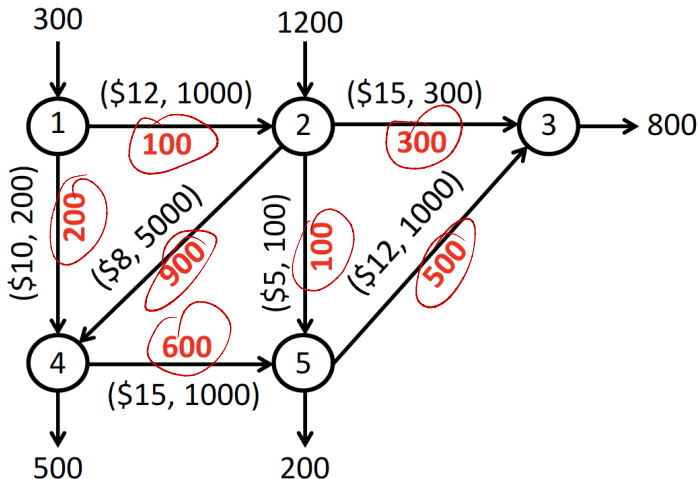
Example Formulation

$$\begin{aligned}
 \min \quad & 12x_{12} + 10x_{14} + 15x_{23} + 8x_{24} \\
 & + 5x_{25} + 15x_{45} + 12x_{53} \\
 \text{s.t.} \quad & x_{12} + x_{14} = 300 \\
 & x_{23} + x_{24} + x_{25} - x_{12} = 1200 \\
 & -x_{23} - x_{53} = -800 \\
 & x_{45} - x_{14} - x_{24} = -500 \\
 & x_{53} - x_{25} - x_{45} = -200 \\
 & 0 \leq x_{12} \leq 1000, \quad 0 \leq x_{14} \leq 200, \\
 & 0 \leq x_{23} \leq 300, \quad 0 \leq x_{24} \leq 5000 \\
 & 0 \leq x_{25} \leq 100, \quad 0 \leq x_{45} \leq 1000 \\
 & 0 \leq x_{53} \leq 1000
 \end{aligned}$$

~~$x_{ij} \in \mathbb{Z}$~~



Example Solution



Integrality of Min Cost Network Flow

Theorem

For any minimum cost network flow problem (LP), if all supplies (b_i) and capacities (u_{ij}) are integers, then the problem has an optimal solution with integer flow on each arc.

then all extreme pts (BFS) are integral.

Totally Unimodular Matrix

Definition

A matrix A is said to be *totally unimodular* (TU) if the determinant of each square submatrix of A is either 0, 1, or -1 .

Examples:


$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \begin{bmatrix} -1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Observation: If A is TU, then each element $a_{ij} \in \{-1, 0, 1\}$.

Theorem

MCNF

In the network flow problem, the matrix of flow conservation constraints is totally unimodular.

One Sufficient Condition for TU

Theorem

Let A be an $m \times n$ matrix. Then the following conditions together are sufficient for A to be totally unimodular:

- ① Every column of A contains at most two non-zero entries;
- ② Every entry in A is 0, +1, or -1;
- ③ The rows of A can be partitioned into two disjoint sets B and C such that
 - (a) If two non-zero entries in a column of A have the same sign, then the row of one entry is in B , and the row of the other in C ;
 $(1, 1)$ or $(-1, -1)$
 - (b) If two non-zero entries in a column of A have opposite signs, then the rows are both in B , or both in C .
 $(1, -1)$

- A 0, +1, -1 matrix A is totally unimodular if it contains no more than one +1 and no more than one -1 in each column.
- A is TU $\iff -A$ is TU $\iff A^T$ is TU.
- A is TU $\iff [A \ I]$ is TU $\iff [A \ -A \ I \ -I]$ is TU.

Example

$$\begin{array}{c}
 \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\
 \text{B} \quad A = \begin{bmatrix} \textcircled{1} & 0 & \textcircled{1} & 0 \\ 0 & \textcircled{1} & 0 & \textcircled{1} \\ \textcircled{1} & \textcircled{1} & 0 & 0 \\ 0 & 0 & \textcircled{1} & \textcircled{1} \end{bmatrix}
 \end{array}$$

Handwritten annotations: A blue circle highlights the top two rows, and a red circle highlights the bottom two rows. Arrows point to each column.

- $B = \{1, 2\}$ and $C = \{3, 4\}$

$$\rightarrow A = \begin{bmatrix} \textcircled{-1} & \textcircled{-1} & 0 & 0 & 0 & \textcircled{1} \\ \textcircled{1} & 0 & \textcircled{-1} & \textcircled{-1} & 0 & 0 \\ 0 & \textcircled{1} & \textcircled{1} & 0 & \textcircled{-1} & 0 \\ 0 & 0 & 0 & \textcircled{1} & \textcircled{1} & \textcircled{-1} \end{bmatrix}$$

Handwritten annotations: A red circle highlights the entire matrix. Arrows point to each column.

- $B = \{1, 2, 3, 4\}$ and $C = \phi$

Properties Totally Unimodular Matrix

Theorem

If the matrix A is TU and the vector b has integer entries, then the polyhedron

$$X = \{x \in \mathbb{R}^n : \overset{\text{TU}}{A}x = \overset{\text{integer}}{b}, x \geq 0\}$$

(if nonempty) has integral extreme points (i.e., each extreme point vector has integral entries).

Cramer's Rule

Corollary

If the matrix A is TU and the vectors a, b, d, f have integer entries, then the following two polyhedra

$$X = \{x \in \mathbb{R}^n : \overset{\text{TU}}{A}x \leq \overset{\text{integer}}{b}\}$$
$$Y = \{x \in \mathbb{R}^n : a \leq Ax \leq b, d \leq x \leq f\}$$

feasible region for MCMF

(if nonempty) have integral extreme points.

Problems Equivalent to Min Cost Network Flow

- Transportation
- Transshipment
- Assignment
- Shortest Path
- Max Flow



Min Cost Network Flow

~~$x_{ij} \in \mathbb{Z}$~~ $\rightarrow 0 \leq x_{ij} \leq u_{ij}$

~~$x_{ij} \in \{0, 1\}$~~ $\rightarrow 0 \leq x_{ij} \leq 1$

NO NEED

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Midterm Exam

- Time and Date: 10 – 12, Sunday, Nov 2
- Location: TC103 / TC206 / TC308 / TC311 / TD307
Please check your assigned location on Blackboard.
- Coverage: everything up to and including Integer Program Modeling and Total Unimodularity.
- Closed-book, Closed-notes, No Cheat Sheet
- You may bring a calculator
- Please bring your student ID to the exam.

- LP Modeling
- IP Modeling
- Simplex Method
- LP Duality Theory
- Sensitivity Analysis

LP Modeling

→ index set, data, variable

- Data-specific/Data-independent

- Production Planning
- Diet Problem
- Transportation
- Scheduling
- Capacity Expansion
- Min-cost Network Flow
- Max Flow

In-class

- Homework 1

Transfer Into LP

- General Principle
- Absolute Value Function
- Piecewise Linear (Maxmin and Minmax)

General Principle

$$\begin{array}{ll} \min. f(x) \\ \text{s.t. } x \in X \end{array} \Leftrightarrow \begin{array}{ll} \min. z \\ \text{s.t. } f(x) \leq z \\ x \in X \end{array}$$

$$\begin{array}{ll} \max. f(x) \\ \text{s.t. } x \in X \end{array} \Leftrightarrow \begin{array}{ll} \max. z \\ \text{s.t. } f(x) \geq z \\ x \in X \end{array}$$

Absolute Value

$$|a^T x + b| \leq z \Rightarrow -z \leq a^T x + b \leq z$$
$$\min. |a^T x + b| \Rightarrow \min. z \text{ s.t. } |a^T x + b| \leq z \Rightarrow \min. z \text{ s.t. } -z \leq a^T x + b \leq z$$
$$|a^T x + b| \geq z \rightarrow \text{Not Possible to transfer to LP}$$

Piece wise linear

$$\min_x \max \{a_1^T x + b_1, \dots, a_m^T x + b_m\}$$
$$\Downarrow$$
$$\min. z$$
$$\text{s.t. } \max_i \{a_i^T x + b_i\} \leq z$$
$$\Downarrow$$
$$\min. z$$
$$\text{s.t. } z \geq a_i^T x + b_i, \forall i$$



$$\max_x \min \{a_1^T x + b_1, \dots, a_m^T x + b_m\}$$



$$\begin{aligned} \max \quad & z \\ \text{s.t.} \quad & \min_i \{a_i^T x + b_i\} \geq z \end{aligned}$$



$$\begin{aligned} \max \quad & z \\ \text{s.t.} \quad & a_i^T x + b_i \geq z, \quad \forall i = 1, \dots, m \end{aligned}$$

- Logic Conditions
- If-Then, Either-Or, K-out-of-N
- Selection
- Facility Location
- Totally Unimodular
- Homework 3

Simplex Method

$$\begin{array}{llll} \text{maximize} & x_1 & +2x_2 & \\ \text{subject to} & x_1 & & \leq 100 \\ & & 2x_2 & \leq 200 \\ & x_1 & +x_2 & \leq 150 \\ & x_1, & x_2 & \geq 0 \end{array}$$

$$\begin{array}{llll}
 \text{maximize} & x_1 & +2x_2 & \\
 \text{subject to} & x_1 & & \leq 100 \\
 & & 2x_2 & \leq 200 \\
 & x_1 & +x_2 & \leq 150 \\
 & x_1, & x_2 & \geq 0
 \end{array}$$

$$\begin{array}{llll}
 \text{min.} & -x_1 - 2x_2 & & = A_B \\
 \text{s.t.} & x_1 & +s_1 & = 100 \\
 & & 2x_2 & +s_2 = 200 \\
 & x_1 + x_2 & & +s_3 = 150
 \end{array}$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

$$X_B = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} = \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix} \quad X_N = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$B = \{3, 4, 5\} \quad A_B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$$X_B = A_B^{-1} b = I^{-1} b = b = \begin{bmatrix} 100 \\ 200 \\ 150 \end{bmatrix}$$

Iteration 1:

Reduced costs

$$\bar{c}_1 = c_1 - \cancel{c_B}^T A_B^{-1} A_1 = c_1 = -1$$

$$\bar{c}_2 = c_2 - \underset{0}{\cancel{c_B}} A_B^T A_2 = c_2 = -2$$

Select $j=1$ to enter the basis.

Basic Directions

$$d_N = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$d_B = \begin{bmatrix} d_3 \\ d_4 \\ d_5 \end{bmatrix} = \begin{bmatrix} ds_1 \\ ds_2 \\ ds_3 \end{bmatrix} = -A_B^T A_1$$

$$= -A_1 = -\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ -1 \end{bmatrix}$$

Min-Ratio Test

$$\theta^* = \min_{i \in B: d_i < 0} \left\{ -\frac{x_i}{d_i} \right\}$$

$$= \left\{ -\frac{s_1}{d_3}, -\frac{s_3}{d_5} \right\}$$

$$= \left\{ -\frac{100}{-1}, -\frac{150}{-1} \right\}$$

$$= \{100, 150\}$$

$$= 100 \quad i=3$$

select $i=3$ to exit basis s_1

$$\begin{bmatrix} x_B \\ x_N \end{bmatrix} + \Theta^* \begin{bmatrix} d_B \\ d_N \end{bmatrix}$$

$$= \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ x_1 \\ x_2 \end{bmatrix} + \Theta^* \begin{bmatrix} d_3 \\ d_4 \\ d_5 \\ d_1 \\ d_2 \end{bmatrix}$$

$$= \begin{bmatrix} 100 \\ 200 \\ 150 \\ 0 \\ 0 \end{bmatrix} + 100 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 200 \\ 50 \\ 100 \\ 0 \end{bmatrix} \begin{matrix} \rightarrow s_1 \\ \rightarrow s_2 \\ \rightarrow s_3 \\ \rightarrow x_1 \\ \rightarrow x_2 \end{matrix}$$

Iteration 2:

$$B = \{4, 5, 1\}$$

$$X_B = \begin{bmatrix} s_2 \\ s_3 \\ x_1 \end{bmatrix} = \begin{bmatrix} 200 \\ 50 \\ 100 \end{bmatrix} \quad X_N = \begin{bmatrix} s_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

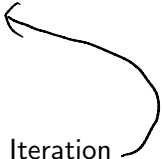
Stopping Criteria:

- $\bar{C} \geq 0$, $\bar{C}_N \geq 0$, x is optimal

- $d \geq 0$, $d_B \geq 0$, LP is unbounded

$$A_B^{-1}$$

Simplex Method

- Standard Form LP
 - Initial Basis and BFS
 - Reduced Costs
 - Basic Direction
 - Min-Ratio Test
 - Update to Next Iteration
 - Stopping Criteria
 - Degeneracy and Two-phase Method
 - Homework 2
- 

$$\rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Degeneracy:

there is zero (s) in basic variables.

Bland's Rule: smallest index

Two-Phase Simplex:

$$\min. e^T y$$

Phase-I LP: s.t. $Ax + y = b$
 $x, y \geq 0$

Initial BFS: $\begin{cases} x=0 \\ y=b \end{cases}$

$\min. C^T x$
s.t. $Ax = b$
 $x \geq 0$

If $y^* \neq 0 \Rightarrow$ original LP is infeasible

If $y^* = 0 \Rightarrow$ original LP is feasible

x^* is a BFS

for original LP

Duality

- Use table to derive the dual.
- Weak and Strong Duality
- Table of Possibles and Impossibles
- Complementary Slackness
- Shadow Price
- Homework 3

$\leq b \leftarrow \text{Resource}$
 (y^*)

Sensitivity Analysis

- Local sensitivity analysis
- Global sensitivity analysis

Basis not change

- If $\Delta \mathbf{b} = \lambda \mathbf{e}_i$, we need

Need
Standard
form
LP

- If $\Delta \mathbf{c} = \lambda \mathbf{e}_j$, we need

- for $j \in N$:

- for $j \in B$:

$$\rightarrow \mathbf{x}_B^* + \lambda \mathbf{A}_B^{-1} \mathbf{e}_i \geq 0 \quad \in \mathbb{R}^m$$

$$\rightarrow \mathbf{r}_N^T - \lambda \mathbf{e}_j^T \mathbf{A}_B^{-1} \mathbf{A}_N \geq 0 \quad \in \mathbb{R}^m$$

$$\rightarrow \mathbf{r}_N + \lambda \mathbf{e}_j \geq 0 \quad \in \mathbb{R}^{n-m}$$

$$\mathbf{r}_N = \mathbf{c}_N - \mathbf{c}_B^T \mathbf{A}_B^{-1} \mathbf{A}_N$$

$$\begin{array}{lll} \text{maximize} & x_1 & + 2x_2 \\ \text{subject to} & x_1 & \leq 100 \\ & & 2x_2 \leq 200 \\ & x_1 & + x_2 \leq 150 \\ & x_1, & x_2 \geq 0 \end{array}$$

Local Sensitivity Analysis

$$b \rightarrow b + \underbrace{\Delta b}_{\text{Small}}, \quad V \rightarrow V + \Delta b^T y^*$$

$$C \rightarrow C + \underbrace{\Delta C}_{\text{Small}}, \quad V \rightarrow V + \Delta C^T x^*$$

(P)

maximize	$x_1 + 2x_2$	
subject to	$x_1 \leq 100$	(y_1)
	$2x_2 \leq 200$	$(y_2) \rightarrow 202$
	$x_1 + x_2 \leq 150$	(y_3)
	$x_1, x_2 \geq 0$	$\rightarrow 150 + \lambda$

min. $100y_1 + 200y_2 + 150y_3$

s.t. $y_1 + y_3 \geq 1 \quad (x_1)$

(D) $2y_2 + y_3 \geq 2 \quad (x_2)$

$y_1, y_2, y_3 \geq 0$

Complementary Slackness

$$y_1 \cdot (x_1 - 100) = 0$$

$$y_2 \cdot (2x_2 - 200) = 0$$

$$y_3 \cdot (x_1 + x_2 - 150) = 0$$

$$x_1 \cdot (y_1 + y_3 - 1) = 0$$

$$x_2 \cdot (2y_2 + y_3 - 2) = 0$$

$$x^* = (50, 100), \quad y^* = (0, 0.5, 1), \quad V = 250$$

RHS of 2nd constraint $\underbrace{200}_{\Delta b=2} \rightarrow 202$

$$V \rightarrow V + \Delta b^T y^*$$

$$V + \Delta b_2 \cdot y_2^*$$

$$250 + \underbrace{(2 \cdot 0.5)}_1 = 251$$

$$\text{min. } -x_1 - 2x_2$$

$$\text{s.t. } x_1 + s_1 = 100$$

$$2x_2 + s_2 = 200$$

$$x_1 + x_2 + s_3 = 150 \rightarrow 150 + \lambda$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

$$x^* = (\overset{x_1^*}{50}, \overset{x_2^*}{100}, \overset{s_1^*}{50}, \overset{s_2^*}{0}, \overset{s_3^*}{0}) \quad \beta = \{1, 2, 3\}$$

$$x_{\beta}^* + \lambda A_{\beta}^T e_i \geq 0$$

$$\begin{matrix} x_1^* \\ x_2^* \\ s_1^* \end{matrix} \leftarrow \begin{bmatrix} 50 \\ 100 \\ 50 \end{bmatrix} + \lambda \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \geq 0$$

$$\Rightarrow -50 \leq \lambda \leq 50$$